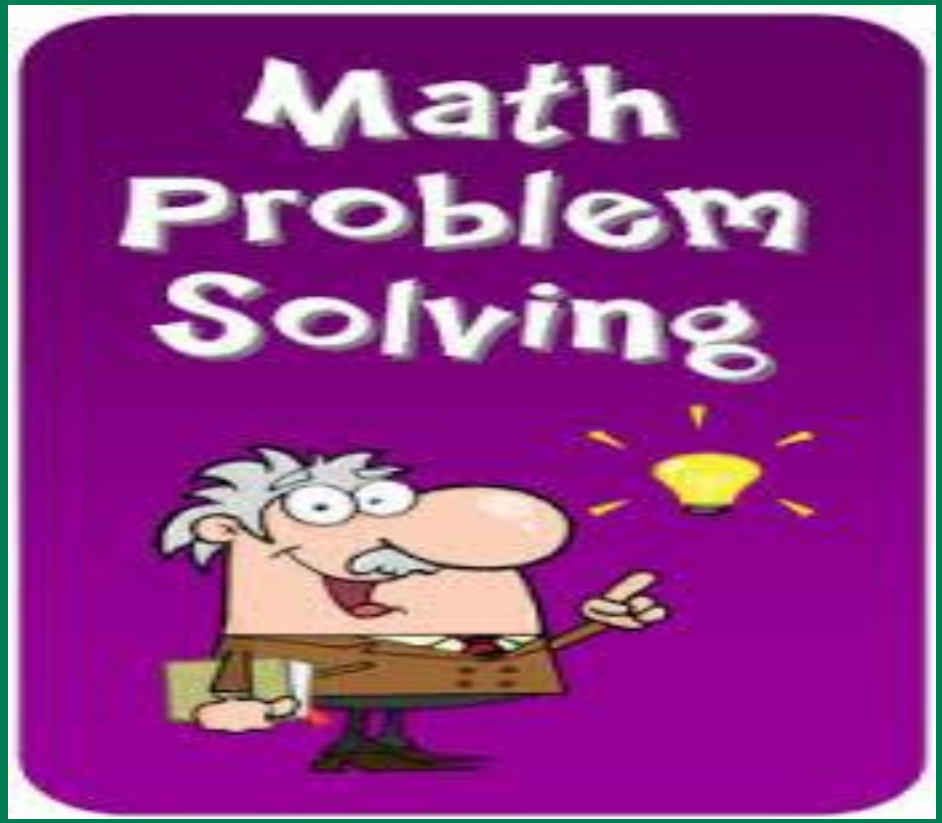




Anthony S. Morfe_CAS_Math Cluster

Objective:

At the end of this presentation, students should be able to solve mathematical problems using different strategies.





Mathematical problem solving is finding a way around a difficulty, around an obstacle, and finding a solution to a problem that is unknown.

George Polya



George Polya (1973) – father of problem solving

Four phases of problem solving

1. Understand the problem
2. Devise a plan
3. Carry out the plan
4. Look back



UNDERSTAND THE PROBLEM

- Look for information given
- Visualize the information
- Organize the information
- Connect the information



DEVISE A PLAN

- To make a representation
  - draw a diagram
  - make a systematic list
  - use equations
- To make a calculated guess
  - guess and check
  - look for a pattern
  - make suppositions



DEVISE A PLAN

- To go through the process
  - act it out
  - work backwards
- To change the problem
  - restate the problem
  - simplify the problem
  - solve part of the problem



CARRYING OUT THE PLAN

- Use mathematical knowledge
- Use mathematical skills
- Use logical thinking



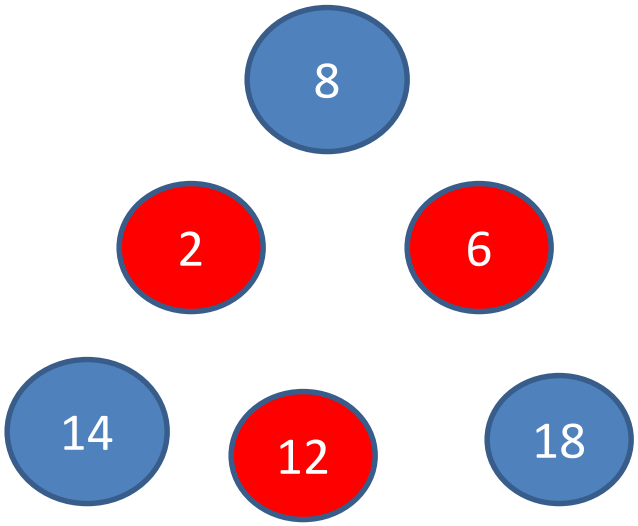
LOOK BACK

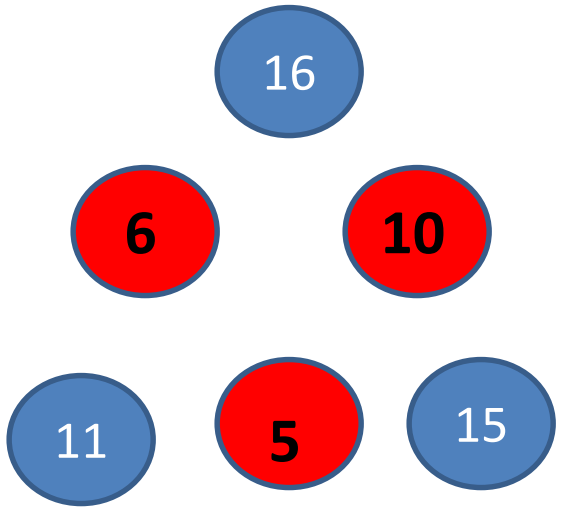
- Check solution
  - Is it reasonable?
- Improve on the method used
- Seek alternative solutions
- Extend the method to other problems



Magic Triangle

In the first diagram, the numbers in the blue circles are found by adding the numbers in the 2 adjacent red circles as shown. Complete the second diagram so that the pattern holds.





Handshakes Problem

In a party there are 5 people. If everyone at the party shakes hands with everyone else, how many handshakes would there be?

Table:

A	B
A	C
A	D
A	E
B	C
B	D
B	E
C	D
C	E
D	E

Answer: 10 handshakes

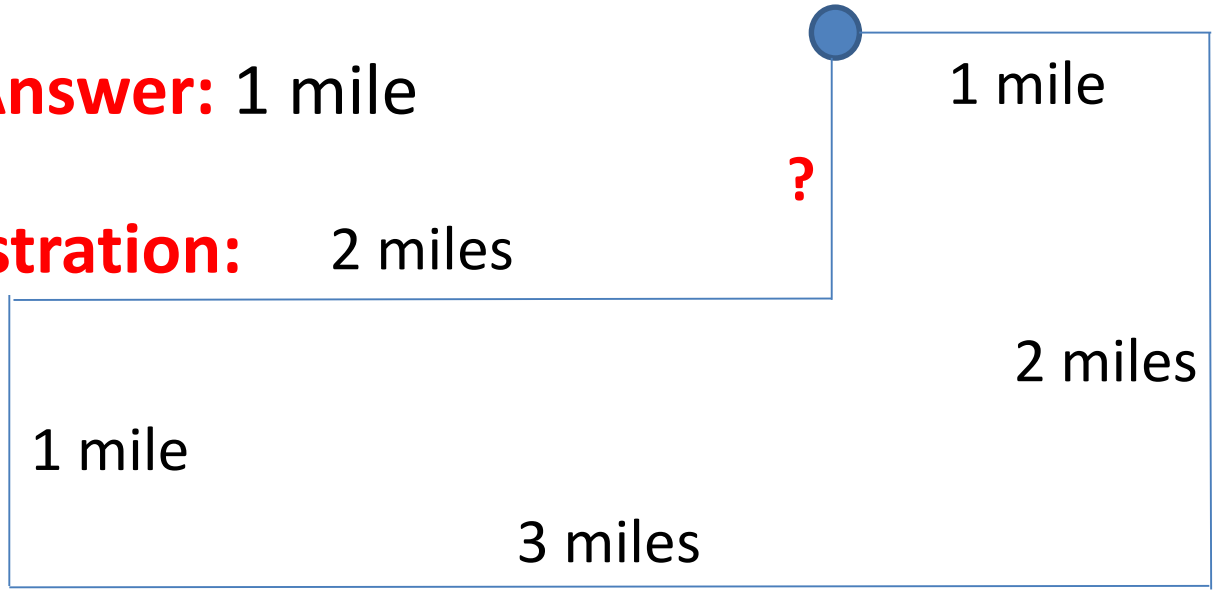


Girl Scout Hike

A Girl Scout troop went on a hike. First they walked 1 mile east, then 2 miles south, then 3 miles west, and 1 mile north, then 2 miles east. How far is the troop from their starting point?

Answer: 1 mile

Illustration:





Draw a Diagram

In a stock car race, the first five finishers in some order were a Ford, a Pontiac, a Chevrolet, a Buick and a Dodge.

(a) The Ford finished 7 seconds before the Chevrolet.

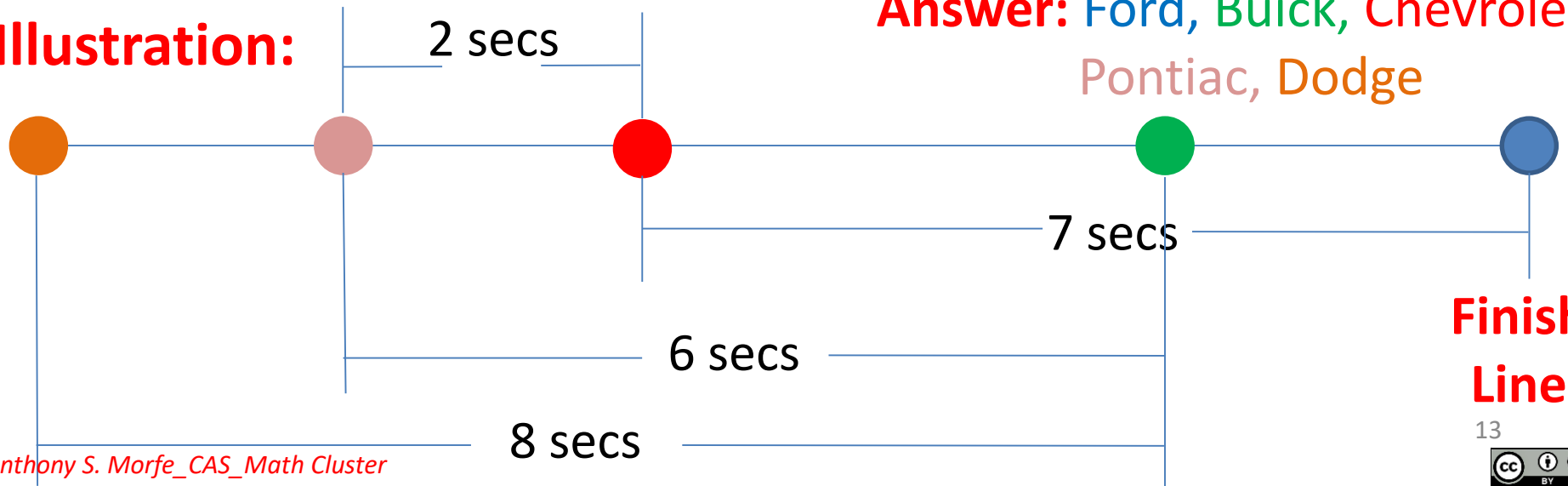
(b) The Pontiac finished 6 seconds after the Buick.

(c) The Dodge finished 8 seconds after the Buick.

(d) The Chevrolet finished 2 seconds before the Pontiac.

In what order did the cars finish the race?

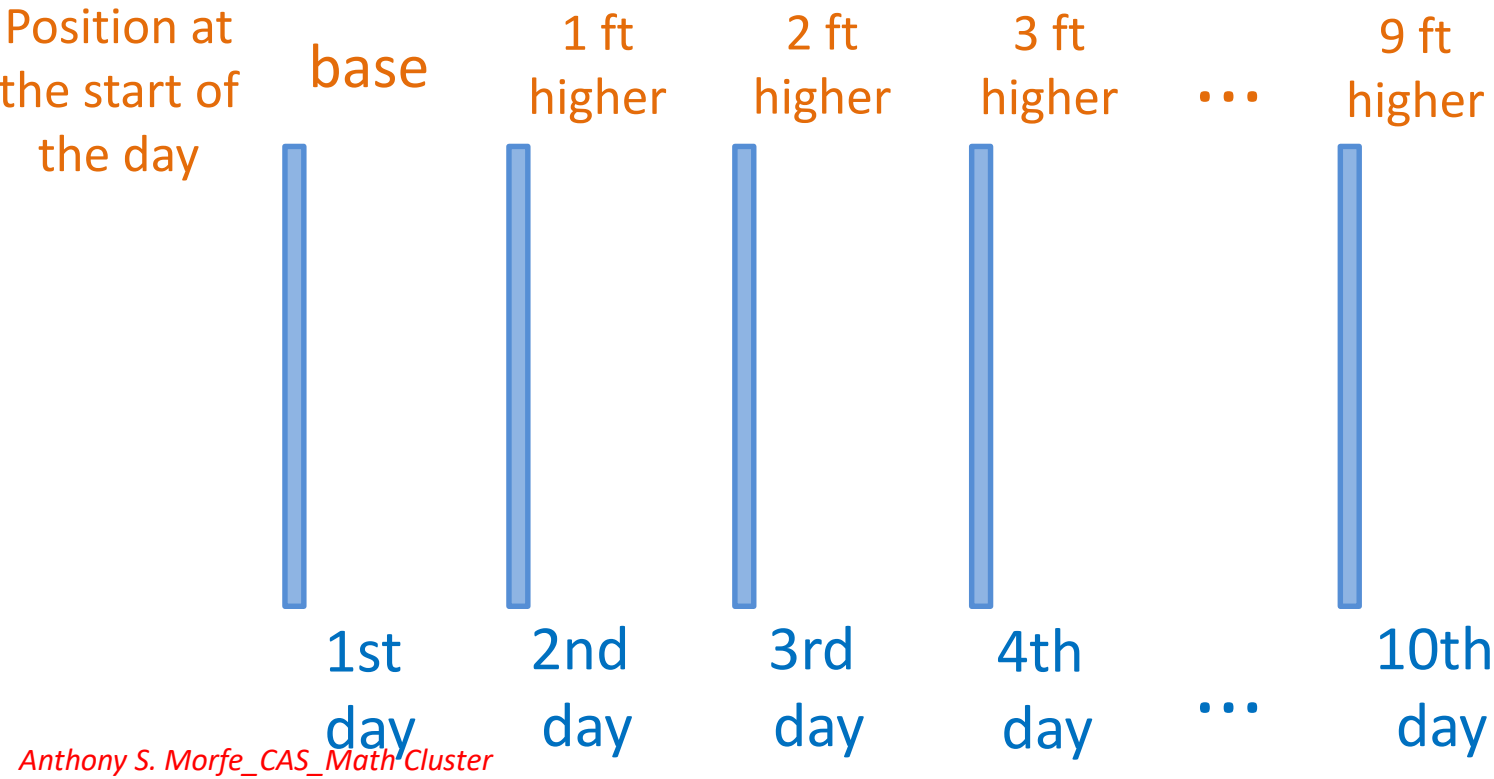
Illustration:

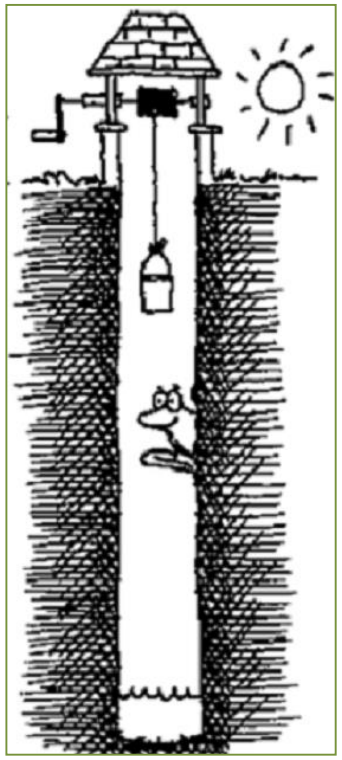


Answer: Ford, Buick, Chevrolet, Pontiac, Dodge

Classic Frog Problem

A frog is in a well 12 feet deep. Each day it climbs up 3 feet and each night it slips back 2 feet. How many days will it take the frog to get out of the well? **Answer:** 10 days





Numerical Sequences

Fill in the next two blanks in each part.

1, 4, 7, 10, 13, ___, ___

16 **19**

Numerical Sequences

Fill in the next two blanks in each part.

19, 20, 22, 25, 29, ___, ___

19, 20, 22, 25, 29, 34, 40

Numerical Sequences

Fill in the next two blanks in each part.

1, 4, 9, 16, 25, 36, 49

Look for a Pattern



What is the sum of the first ten counting numbers?

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = ?$$

Answer: 55

Look for a Pattern

What is the sum of the first one hundred counting numbers?



$$1 + 2 + 3 + 4 + 5 + 6 + \ldots + 95 + 96 + 97 + 98 + 99 + 100 = ?$$

Answer: 5050

Use a Variable

In order to encourage his son in the study of Algebra, a father promised the son P8 for every problem solved correctly and to fine him P5 for each incorrect solution. After 26 problems neither owed anything to the other. How many problems did the boy solve correctly?

Solution: Let x be the number of questions that the son has solved correctly.

$$8x - 5(26 - x) = 0$$

$$8x - 130 + 5x = 0$$

$$13x = 130$$

$$x = 10$$

Eliminate Possibilities

Elyra, Ejah, Wino, Angel and Wenet are aliens on a space ship.

(a) Ejah is younger than Angel.

(b) Ejah is not the youngest in the group.

(c) Only one alien is older than Wenet.

(d) Wenet is younger than Wino.

Arrange Elyra, Ejah, Wino, Angel and Wenet in order of increasing age.

Answer:

Elyra	Ejah	Angel	Wenet	Wino
Youngest				Oldest

Work Backwards

While three watchmen were guarding an orchard, a thief slipped in and stole some apples. On his way out, he met the three watchmen, one after the other. To each in turn, he gave half of the apples he then had, and two more. In this way he managed to escape with one apple. How many apples did the thief steal originally?

Solution:

Number of apples at hand before meeting the 3rd watchman: $(1 + 2) \times 2 = 6$ apples

Number of apples at hand before meeting the 2nd watchman: $(6 + 2) \times 2 = 16$ apples

Number of apples at hand before meeting the 1st watchman: $(16 + 2) \times 2 = 36$ apples

Answer: 36 apples

Work Backwards

By 7 pm all the party guests had arrived but by 8 pm a third of them had departed. By 9 pm a third of those remaining had also departed. By 10 pm the same happened again and a third of those remaining had also departed. Finally the party ended at 11 pm by which time further third of those remaining had also departed and by 11 pm only 16 guests still remained. How many guests were originally at the party at 7 pm?

Solution: Number of guests by 11pm: 16 guests

Number of guests by 10pm: (16 x 3)/2 = 24 guests

Number of guests by 9pm: (24 x 3)/2 = 36 guests

Number of guests by 8pm: (36 x 3)/2 = 54 guests

Number of guests by 7pm: (54 x 3)/2 = 81 guests

Answer: 81 guests

Reference

- Polya, G. (1957). How to Solve it: A New Aspect of Mathematical Method.USA: Princeton University Press.



Thank You!!!



